

HU Collection

TMUA Mock Set I Paper 2

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	A	C	F	D	B	E	E	D	A	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	E	D	B	F	D	C	B	A	A	E

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer A.

Independence means $P(X \cap Y) = P(X)P(Y)$. I: $P(A \cap B) = \frac{3}{36} = \frac{1}{2} \cdot \frac{1}{6}$, independent. II: $P(A \cap C) = \frac{2}{36}$, from $(2, 4)$ and $(4, 2)$, but $P(A)P(C) = \frac{1}{2} \cdot \frac{5}{36}$. III: D is contained in A , so $P(A \cap D) = \frac{1}{4} \neq \frac{1}{8}$. IV: two even numbers never total 7, so $P(B \cap D) = 0$, while $P(B)P(D) > 0$. Only I. It is striking that a total of 7 carries no information about the first die, but a total of 6 does.

Question 2. Answer C.

The definition asks that f eventually stays above any level, not merely that it gets large sometimes. I: $x - \sin x \geq x - 1$, so $N = M + 1$ works. V: $x + 10 \sin x \geq x - 10$, so $N = M + 10$ works, despite the large oscillation. III: $(\ln x)^2 > M$ once $x > e^{\sqrt{M}}$, slowly but surely. II fails: $x \sin x$ reaches arbitrarily large values but returns to 0 at every multiple of π , so it never stays above $M = 1$. IV fails: $xe^{-x} \rightarrow 0$. Three functions.

Question 3. Answer F.

The graph is symmetric about $x = 1$. I: reflecting, $\int_0^2 xf = \int_0^2 (2-x)f$, so $2 \int_0^2 xf = 2 \int_0^2 f = 12$. II: $(x-1)^3$ is odd about $x = 1$ and f is even about it, so the integral is 0. IV: differentiating $f(x) = f(2-x)$ gives $f'(x) = -f'(2-x)$, and $x = 1$ gives $f'(1) = 0$. III is not determined: $f = 3$ gives 8, but $f(x) = 2 + 3(x-1)^2$, which also has integral 6, gives $\frac{128}{15}$.

Question 4. Answer D.

Cases I and II are right, the four cases cover everything, and IV is a legitimate symmetry. The gap is in III. When $a \geq 0 > b$, the number $a + b$ may be negative, and then $|a + b| = -(a + b)$. The inequality $a + b < a$ bounds $a + b$ from above but says nothing about $-(a + b)$. The repair is one line: $-(a + b) = -a - b \leq -b = |b| \leq |a| + |b|$. So the claim is true, and the proof is incomplete.

Question 5. Answer B.

I: by AM-GM, $u_{n+1} \geq \sqrt{u_n \cdot \frac{4}{u_n}} = 2$, with equality only if $u_n = 2$. Since $u_2 = \frac{5}{2}$, the terms never equal 2, so $u_n > 2$ from $n = 2$ on. II: $u_{n+1} - u_n = \frac{4 - u_n^2}{2u_n} < 0$ when $u_n > 2$. III: rational operations

on rationals. IV fails for the same reason as I: the terms approach 2 very quickly ($u_5 \approx 2.0000001$) but never reach it.

Question 6. Answer E.

I: $p \rightarrow -\infty$ on the left and $p(-1) > 0$ give a root below -1 ; $p(-1) > 0 > p(1)$ gives one in $(-1, 1)$; and $p(1) < 0$ with $p \rightarrow +\infty$ gives one above 1. III: $p = x(x^2 + ax + b)$, and the quadratic has discriminant $a^2 - 4b > 0$ when $b < 0$. II guarantees the opposite: then $p' = 3x^2 + 2ax + b > 0$ everywhere, so there is exactly one real root. IV gives $x^3 + c$, which has one real root unless $c = 0$.

Question 7. Answer E.

I: $e^x - 1 - x$ has derivative $e^x - 1$, so its minimum is at $x = 0$, with value 0: the line $y = 1 + x$ is the tangent at $(0, 1)$ and touches exactly once. III is the same picture for the inverse function: $y = x - 1$ is the tangent to $\ln x$ at $(1, 0)$, and $\ln x$ lies below it everywhere else. II has no solution: $e^x - 2x$ has minimum value $2 - 2 \ln 2 > 0$ at $x = \ln 2$. IV has three: $x = 2$, $x = 4$, and one negative solution between -1 and 0, where 2^x is still positive and x^2 falls to 0.

Question 8. Answer D.

With $n = r + b$: $P(\text{same}) = \frac{r(r-1)+b(b-1)}{n(n-1)}$ and $P(\text{different}) = \frac{2rb}{n(n-1)}$. I: equating gives $r^2 + b^2 - r - b = 2rb$, that is $(r - b)^2 = r + b$. II: $r = b$ gives $\frac{2r^2}{2r(2r-1)} = \frac{r}{2r-1} > \frac{1}{2}$. III: $r = 3$, $b = 1$ gives $\frac{3 \cdot 2}{4 \cdot 3} = \frac{1}{2}$. IV fails for $r = b = 1$, where the balls are certainly different colours.

Question 9. Answer A.

I: an increasing function is determined by its set of three values, so $\binom{4}{3} = 4$. II: a non-decreasing function is a choice of three values with repetition allowed, arranged in order, so $\binom{4+3-1}{3} = 20$. III: of the $3^4 = 81$ functions, remove those missing a value: $3 \cdot 2^4 = 48$ miss at least one, but the 3 constant functions have been removed twice, so add them back: $81 - 48 + 3 = 36$. IV fails: there are $4 \cdot 3 \cdot 2 = 24$; the figure 12 comes from wrongly dividing by 2.

Question 10. Answer C.

I: substituting $b = \frac{a+c}{2}$, $\cos B = \frac{a^2+c^2-b^2}{2ac} = \frac{3a^2+3c^2-2ac}{8ac} \geq \frac{6ac-2ac}{8ac} = \frac{1}{2}$. III: the triangle inequality $a + b > c$ becomes $a + \frac{a+c}{2} > c$, that is $3a > c$. II fails for 3, 4, 5. IV fails too: 2, 3.5, 5 is an arithmetic progression and a valid triangle, but $5^2 > 2^2 + 3.5^2$, so C is obtuse. So I and III.

Question 11. Answer E.

The sum is even when both numbers have the same parity. If $n = 2m$, there are m of each parity, and the probability is $\frac{2\binom{m}{2}}{\binom{2m}{2}} = \frac{m-1}{2m-1}$; setting this to $\frac{7}{15}$ gives $m = 8$, so $n = 16$. If $n = 2m + 1$, there are $m + 1$ odd and m even numbers, and the probability is $\frac{\binom{m+1}{2} + \binom{m}{2}}{\binom{2m+1}{2}} = \frac{m}{2m+1}$; this gives $m = 7$, so $n = 15$. Both work, and only the two-case analysis finds both.

Question 12. Answer D.

I: the integrand is positive on $(0, \pi)$. II: on $(\pi, 2\pi)$, $\sin x$ is negative and the weight x is larger there than on $(0, \pi)$, so the negative half wins (in fact the value is -2π). III: the integrand is odd, so the integral is 0, not positive. IV: $\sin(x + \pi) = -\sin x$ pairs each point of the positive half with one of

the negative half, and the weight $\frac{1}{1+x}$ is larger on the positive half, so the integral is positive. Here the decreasing weight favours the first hump, where in II the increasing weight favoured the second.

Question 13. Answer B.

The function $g(x) = f(x) - x$ has $g' \leq 0$, so g is non-increasing on $[0, 1]$. But $g(0) = 0 = g(1)$, so g is constant, equal to 0: this is I. III and IV follow from I. II is not forced, because the condition says nothing about f outside $[0, 1]$: the function equal to x on $[0, 1]$ and to $x + (x - 1)^2 \cdot 5$ beyond it is differentiable and has $f(2) = 7$. The squeeze, a slope of at most 1 over a rise of exactly 1, leaves no room at all.

Question 14. Answer F.

I: if $f(0) < 0$, then f is negative just to the right of 0, making $F(t) < 0$ for small t . II: if $f(2) > 0$, then f is positive just to the left of 2, so $F(t) = -\int_t^2 f < 0$ for t close to 2. III: $\int_0^2 f = 0$, so f cannot keep one sign throughout, and by continuity it has a zero. IV fails: $F(t) = t^2(2 - t)$ is non-negative on $[0, 2]$ and vanishes at both ends, but $f(x) = F'(x) = 4x - 3x^2$ has $f(1) = 1$. The zero guaranteed by III sits wherever F peaks, which here is at $x = \frac{4}{3}$, not at the midpoint.

Question 15. Answer D.

I: any triple $(-k, 0, k)$ with $k \geq 1$ works, since both sides are 0. II: with $1 \leq x < y < z$, the sum is less than $3z$, so $xy < 3$, forcing $x = 1$, $y = 2$, and then $3 + z = 2z$ gives $z = 3$. IV: if all are non-zero and of one sign, the triple is $(1, 2, 3)$ or, negating everything, $(-3, -2, -1)$. Mixed signs are impossible: if exactly one is negative, say $x < 0 < y < z$, the right side is negative, so $|x| > y + z$ and the left side has size $|x| - y - z < |x|$, while the right side has size $|x|yz \geq 2|x|$. Two negatives reduce to this case by changing every sign. III fails: $(-3, -2, -1)$ has product -6 .

Question 16. Answer C.

$f'(x) = x(4x^2 + 3ax + 2b)$, so $x = 0$ is always stationary, and the question is what the quadratic adds. II: when $9a^2 < 32b$ the quadratic has no real roots, so $x = 0$ is the only stationary point. I is the special case $a = 0$, $b > 0$. IV: $f = x^4$, whose only stationary point is 0. III fails: $a = 5$, $b = 1$ has $9a^2 > 32b$, and f' has three roots. (The complete answer is $9a^2 < 32b$, or $a = b = 0$.)

Question 17. Answer B.

$f'' > 0$ means the graph bends upwards: it lies strictly below its chords and above its tangents. I: $(1, f(1))$ is below the chord from $(0, 0)$ to $(2, f(2))$, so $f(1) < \frac{f(2)}{2}$. II: $(0, 0)$ is below the chord from $(-1, f(-1))$ to $(1, f(1))$, so $0 < \frac{f(1)+f(-1)}{2}$. III: the graph lies above its tangent at the origin, $y = f'(0)x$. IV fails: $f(x) = x^2 - 3x$ has $f'' = 2 > 0$ and $f(0) = 0$, but $f'(1) = -1$.

Question 18. Answer A.

$g(x) = x|x| + px$ is odd. I: $x|x|$ is increasing and so is px when $p \geq 0$, so g is increasing and meets each level once. II: $x(|x| + p) = 0$ gives $x = 0$ or $|x| = -p$, three solutions. III: for $p < 0$, g has a local maximum $\frac{p^2}{4}$ at $x = \frac{p}{2}$ and a local minimum $-\frac{p^2}{4}$ at $x = -\frac{p}{2}$, so $q = \mp \frac{p^2}{4}$ gives a tangency and exactly two solutions. IV fails: each half-line $x \geq 0$ and $x < 0$ carries a quadratic, and the shape of g (rising, falling, rising) allows at most three crossings of any horizontal line.

Question 19. Answer A.

II implies I, since differentiable functions are continuous. For II, take $u < v$. If both lie on the same side of 0, the mean value theorem on $[u, v]$ gives $f(v) - f(u) = f'(c)(v - u) > 0$. If $u < 0 < v$, apply it on $[u, 0]$ and on $[0, v]$, which needs continuity at 0, and add. III fails without continuity: $f(x) = x$ for $x < 0$ and $f(x) = x - 1$ for $x \geq 0$ has $f' = 1$ away from 0, but $f(-0.1) > f(0)$. IV fails: x^3 is increasing, with $f'(0) = 0$.

Question 20. Answer E.

A solution needs $x + b \geq 0$ and then, squaring, $x + a = (x + b)^2$, that is $x^2 + (2b - 1)x + b^2 - a = 0$. (The condition $x + a \geq 0$ follows automatically from the squared equation.) Two distinct solutions need two distinct roots of this quadratic, both at least $-b$. Distinct roots: $(2b - 1)^2 - 4(b^2 - a) = 1 - 4b + 4a > 0$, that is $a > b - \frac{1}{4}$. The vertex $x = \frac{1-2b}{2}$ always exceeds $-b$, so both roots are at least $-b$ exactly when the quadratic is non-negative at $x = -b$: $b^2 - (2b - 1)b + b^2 - a = b - a \geq 0$. So $b - \frac{1}{4} < a \leq b$. At $a = b$ the root $x = -b$ is included (the right side is 0), which is why that end is closed.